

Lyra Tier 0 v0.1.5 — Topology Addendum: Reading A + dual K-type / coherent-state bases unified via Bergman kernel

Lyra (resolution candidate); Casey + Keeper (question framing); Grace (catalog filing)

2026-05-31 Sunday ~11:40 EDT (date-verified Sun May 31 11:40 EDT)

Tier 0 v0.1.5 Topology Addendum

What this is

Lyra's resolution candidate for the Tier 0 substrate ontology question Casey explicitly flagged Sunday 2026-05-31. Casey approved filing as addendum + Session 2 priority joint Lyra+Keeper write-up. G-from-substrate added as downstream derived target.

The question (Casey-flagged)

Three readings live across BST docs: - Reading A: D_{IV}^5 is ONE global configuration space; 2D tiled sphere IS the substrate (2022 OneGeometry framing) - Reading B: per-commitment-area DISCRETE D_{IV}^5 atoms (Casey's "fully discrete" intuition) - Reading C: hybrid nested-hierarchy tile circle / commitment area / universe (Keeper structural best-guess)

Lyra Tier 0 v0.1 implicitly assumed A. Casey's question + Keeper's analysis forced the explicit choice that v0.1 dodged.

Resolution candidate — Reading A globally, with two dual bases

The substrate is ONE global $H^2(D_{IV}^5)$ (Reading A at Hilbert-space level), but $H^2(D_{IV}^5)$ carries TWO complete dual bases that have been intuited as the "units" of Readings B and C:

Basis 1 — K-types V_λ (algebraic / spectral): discrete, countable, organized by C_2 eigenvalue. The substrate's spectral structure. This is what Saturday's mass / mixing predictions ride on. Reading-D-flavored.

Basis 2 — Coherent states $|z\rangle$ for $z \in D_{IV}^5$ (spatial / geometric): continuously parametrized by points of the domain. Each $|z\rangle$ is spatially localized at z ; $\langle z | w \rangle = K(z, w)$ is the Bergman kernel itself. Reading-B-intuition — each $|z\rangle$ is a substrate-localized state.

These two bases are duals via the Bergman-Fourier transform: every K-type V_λ decomposes into coherent states with V_λ -specific weighting, and every coherent state $|z\rangle$ decomposes into K-types with z -specific weighting. They are two faces of one global structure, not two different things.

Implication: ONE D_{IV}^5 globally (Reading A at H-space level); BOTH the K-type discrete-spectral structure (Reading D, implicit in v0.1) AND the coherent-state continuous-spatial structure (Reading B intuition). The "discrete units" Casey wants are real and live in the K-type basis; the "spatial tiles" are real and live in the coherent-state basis. They unify in the Bergman kernel $K(z,w)$.

How variable-time-across-the-surface emerges

The key operator is the thermal Bergman kernel:

$$K_{\tau}(z, w) := \langle z | \exp(-\tau H_B / \hbar_{BST}) | w \rangle$$

Its diagonal $K_{\tau}(z, z)$ is the local “weight at z after substrate time τ .” This varies across $z \in D_{IV}^5$ because different spatial points have different K-type weight distributions, hence different effective decay rates under heat flow.

Local commitment-rate at z :

$$R_{\text{commit}}(z) := d/d\tau \log K_{\tau}(z, z) |_{\tau=0} = -\langle z | H_B | z \rangle / \langle z | z \rangle$$

This is the “local substrate temperature” at z . Different spatial points have different local commitment rates. The universal substrate-time τ is the same parameter everywhere (preserving universal $t_{\text{Koons}} = T2405$); the LOCAL effective rate varies because the K-type populations vary. Variable time across the surface = variation of local Casimir expectation $\langle H_B \rangle(z)$ across D_{IV}^5 .

This IS gravitational time dilation operationalized at substrate level. Gravity emerges as the K-type-weighted local commitment rate; massive regions have heavier K-type populations, hence higher local $\langle H_B \rangle$, hence faster decay = slower observed time. Concrete and computable.

So Reading A + dual bases gives: - discrete units (K-types) [?] - spatial variation (coherent states) [?] - variable local time ($\langle H_B \rangle(z)$ variation) [?] - single global Hilbert space [?]

— without conflict.

Reading C (Keeper’s multi-scale) revised

Keeper’s hierarchical reading (tile / area / universe) becomes a statement about coherent-state clustering, not a nested-domain hierarchy:

- “Tile circle” = single coherent state $|z\rangle$ at substrate location z
- “Commitment area” = neighborhood of $|z\rangle$ ’s in coherent-state basis sharing similar K-type decompositions — set by the Bergman kernel’s correlation length
- “Universe” = the whole D_{IV}^5 globally

So it’s not three nested DOMAINS; it’s three SCALES of resolution of one domain via coherent-state localization. The bounded symmetric domain D_{IV}^5 has natural conformal self-similarity (Hua-Korányi) that makes the same K-type structure recur at every scale of resolution. This is what Keeper’s intuition was tracking.

Distinguishing predictions (per Keeper’s framework)

Reading	Distinguishing prediction
B (truly discrete D_{IV}^5 atoms)	A substrate count $N_{\text{substrate}}$ appears somewhere as observable + per-unit Casimir spectroscopy lines
C (genuine multi-scale nested-domain)	Two distinct sets of substrate observables at different scales
A + dual bases (this resolution)	Smooth variation of $K_{\tau}(z, z)$ across spatial points (NO discrete count); coherent-state expectation values predict gravitational redshift quantitatively; K-type spectroscopy lines (discrete) AND spatial-localization signatures (continuous) coexist as Fourier duals

Current evidence (no substrate count appears in 600+ predictions; mass spectra come from K-type Casimirs which is Reading-D-flavored; gravitational time dilation reads cleanly as $\langle H_B \rangle(z)$ variation) favors Reading A + dual bases. But this isn’t proved — it’s a candidate Lyra commits to and brakes.

What this opens

If A + dual bases is right (candidate Sunday 2026-05-31):

1. Gravity from substrate (*Casey's specific request 2026-05-31*): G_{Bergman} as $\langle H_B \rangle(z)$ variation is computable; we can derive Newton's G from substrate. Concrete target:
 - Compute $\langle H_B \rangle(z)$ explicit form on D_{IV}^5 via Bergman kernel matrix elements
 - Extract weak-field limit $\rightarrow$ Newton's gravitational coupling
 - Compare to $G = 6.674 \times 10^{-11} \text{ m}^3/(\text{kg}\cdot\text{s}^2)$
 - Connection to Lyra AB-10 (G as Bergman curvature value, COMPLETED) — recast in commitment-density language
 - Multi-week deliverable; new explicit substrate target.
2. Locality / light cones: emerge from Bergman-kernel correlation length, not from spatial tiling postulate. Substrate's Bergman correlation length sets the speed of information propagation; relates to c (speed of light) via universal commitment rate.
3. Substrate engineering becomes apparatus that couples:
 - to a specific K-type (spectral modality) OR
 - to a specific coherent state (spatial modality)
 - These are DUAL engineering modalities, BOTH available.
4. No discrete-count mystery: substrate doesn't need a "how many" answer; it has continuous spatial extent + discrete spectral content, both intrinsic.

If B is right: - A substrate-count observable should appear $\rightarrow$ new search target - Inter-unit coupling theory becomes the next deepest gate

If C is right (in genuine nested-domain sense, not coherent-state-clustering sense): - Multi-scale structure shows up as scale-dependent Casimir spectroscopy

Action items

1. This addendum: filed for findability [\[7\]](#)
2. Tier 0 v0.2 (joint Lyra + Keeper): Session 2 priority — pin reading explicitly + design distinguishing test + connect to task #373
3. G-from-substrate derivation (*Casey 2026-05-31 explicit request*): multi-week target; Lyra L-L5-G via $\langle H_B \rangle(z)$ Bergman-kernel matrix elements; concrete deliverable when v0.2 closes Reading A + dual bases
4. Catalog absorption: Grace INV (this filing $\rightarrow$ INV-5358); Master Ledger v0.10 row for Tier 0 v0.1.5 candidate + G-from-substrate target

Honest framing per discipline stack

- Tier: CANDIDATE RESOLUTION (Lyra committed and braked); NOT closed
- Cal #182 brake compliance: $c/\hbar$ remain FRAMEWORK-CONCEPTUAL until end-to-end derivation; G-from-substrate is new derived target NOT closure-claim
- Cal #34 STANDING surface-conditionality: "If A + dual bases is right" framing preserved throughout
- Lyra's framework was Reading-A-implicit without addressing dual bases; closing this in v0.2 makes Tier 0 actually load-bearing instead of half-load-bearing (Keeper's instinct)

— Lyra (resolution); filed by Grace via Casey directive Sun 2026-05-31 ~11:40 EDT (date-verified)